CO1 - AT2

ITA0329 - Mobile Computing — Galla Sai Siddartha

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Q1.

Given:

Total bandwidth = 45 MHz = 45000 kHz

Simplex channel bandwidth = 25 kHz

Formula:

Number of full-duplex channels = Total bandwidth / (2 × Channel bandwidth)

= 45000 / (2 × 25)

= 45000 / 50 = 900

Answer: 900

Q2.

Given:

Total channels = 560

Cluster size = 7

Formula : Channels per cell = Total channels / Cluster size

= 560 / 7 = 80

Answer: 80 channels.

Q3.

Given:

Number of carriers = 100

Time slots per carrier = 8

Formula : Total voice channels = Number of carriers × Time slots

= 100 × 8 = 800

Answer: 800 channels.

Q4.

Given:

Base station power = 80 W

Handset power = 4 W

Formula : Power ratio = 10 log10 (P_1 / P_2)

= 10 log10(80 / 4)

= 10 log10 (20) = 13.01 dB

Answer: The power ratio is 13.01 dB.

Q5.

Given:

Carrier power = −60 dBm

Interference power = −92 dBm

Formula : C/I = Carrier power − Interference power

= −60 − (−92)

= −60 + 92 = 32 dB

Answer: The C/I ratio is 32 dB.

Q6.

Given:

Total bandwidth = 42 MHz = 42000 kHz

Channel bandwidth = 300 kHz

Cluster size = 7

Formula:

Total channels = Total bandwidth / Channel bandwidth

Channels per cell = Total channels / Cluster size

Total channels = 42000 / 300 = 140

Channels per cell = 140 / 7 = 20

Answer: Each cell gets 20 channels.

Q7.

Given:

Initial received power = 64 W

Distance ratio = 2

Path-loss exponent = 4

Formula : $P_2 = P_1$ / (Distance ratio)^2

P_2 = 64 / 2^2

= 64 / 16 = 4 W

Answer: received power is 4 W.

Q8.

Given:

Original radius = R

New radius = 2R

Path-loss exponent = 2

Formula : $P_2 / P_1 = (R_2 / R_1)^2$

$= (2R / R)^2$

$= 2^2 = 4$

Answer: 4 times.

Q9.

Given:

Number of cells = 4

Channels per cell = 15

Formula : Total channels = Number of cells × Channels per cell

$= 4 \times 15 = 60$

Answer: 60 channels.

Q10.

Given:

Vehicle speed = 100 km/h

Carrier frequency = 900 MHz

Speed of light = 3×10^8 m/s

Formula : Doppler shift = (Velocity × Frequency) / Speed of light

Velocity = 100 / 3.6 = 27.78 m/s

Doppler shift = $(27.78 \times 900 \times 10^6) / (3 \times 10^8) = 83.33$ Hz

Answer: 83.33 Hz.

Q11.

Given:

Frame duration = 5 ms

Number of time slots = 8

Formula : Slot duration = Frame duration / Number of slots

$= 5 / 8 = 0.625$ ms

Answer: 0.625 ms.

Q12.

Given:

Total bandwidth = 24 MHz = 24000 kHz

Carrier spacing = 200 kHz

Formula : Number of carriers = Total bandwidth / Carrier spacing

$= 24000 / 200 = 120$

Answer: 120 carriers

Q13.

Given:

Transmitted power = 8 W = 8000 mW

Path loss = 33 dB

Formula:

Transmitted power (dBm) = $10 \log_{10}$ (Power in mW)

Received power = Transmitted power − Path loss

Transmitted power = $10 \log_{10}$ (8000) = 39.03 dBm

Received power = 39.03 − 33 = 6.03 dBm

Answer: The received power is 6.03 dBm.

Q14.

Given:

Frequency = 1900 MHz

Distance = 2.5 km

Formula : FSPL = $32.44 + 20 \log_{10}(f) + 20 \log_{10}(d)$

= $32.44 + 20 \log_{10}(1900) + 20 \log_{10}(2.5)$

= 32.44 + 65.58 + 7.96

= 105.98 dB

Answer: 106 dB.

Q15.

Given:

Number of cells = 6

Calls per cell = 250

Formula : Total capacity = Number of cells × Calls per cell

= 6 × 250 = 1500

Answer: 1500

Q16.

Given:

Call arrival rate = 4000 calls/second

Average call duration = 3 minutes = 180 seconds

Formula : Active calls = Call arrival rate × Average call duration

= 4000 × 180

= 720000

Answer: 720000 active calls

Q17.

Given:

Number of channels = 150

Bandwidth per channel = 25 kHz

Formula:

Total bandwidth = Number of channels × Bandwidth per channel

= 150 × 25 = 3750 kHz

= 3.75 MHz

Answer: bandwidth consumed is 3.75 MHz.

Q18.

Given:

Number of frequency bands = 30

Guard-band width = 40 kHz

Formula:

Number of guard bands = Number of bands − 1

Total guard-band bandwidth = Number of guard bands × Guard-band width

Number of guard bands = 30 − 1 = 29

Total bandwidth = 29 × 40 = 1160 kHz

Answer: The total guard-band bandwidth is 1160 kHz or 1.16 MHz.

Q19.

Given:

Vehicle speed = 90 km/h

Cell radius = 1.5 km

Formula:

Cell diameter = 2 × Radius

Handoffs per hour = Distance travelled / Cell diameter

Cell diameter = 2 × 1.5 = 3 km

Handoffs = 90 / 3 = 30

Answer: 30 handoffs .

Q20.

Given:

Regional centers = 8

Base stations per center = 40

Users per base station = 250

Formula : Total capacity = Centers × Base stations × Users per base station

= 8 × 40 × 250

= 80000

Answer: 80000 users.

Q21.

Given:

Total area = 400 km²

Cell radius = 400 m = 0.4 km

Formula:

Cell area = R^2

Number of cells = Total area / Cell area

Cell area = $(0.4)^2 = 0.16$ km²

Number of cells = 400 / 0.16 = 2500

Answer: 2500 cells

Q22.

Given:

Total spectrum = 6 MHz = 6000 kHz

Bandwidth per channel = 30 kHz

Formula : Number of channels = Total bandwidth / Bandwidth per channel

= 6000 / 30 = 200

Answer: 200 channels

Q23.

Given:

Number of subscribers = 1200

Calls per hour per subscriber = 4

Average call duration = 2.5 minutes

Formula : Traffic = (Subscribers × Calls per hour × Call duration) / 60

= (1200 × 4 × 2.5) / 60

= 12000 / 60 = 200 Erlangs

Answer: 200 Erlangs.

Q24.

Given:

Total bandwidth = 40 MHz = 40000 kHz

Bandwidth per user = 400 kHz

Formula : Maximum users = Total bandwidth / Bandwidth per user

= 40000 / 400 = 100

Answer: 100 users.

Q25.

Given:

Number of base stations = 1200

Users per base station = 600

Formula : Total capacity = Number of base stations × Users per base station

= 1200 × 600 = 720000

Answer: 720000 users.